Lab06-Approximation

CS2312-Algorithm and complexity, Xiaofeng Gao, Spring 2026.

$*$ Contact TA Yixuan Cai for any questions. Include both your .pdf and .tex files in the uploaded .rar or .zip file.

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1. (Bin Packing) We are given a list of real numbers $a_1, a_2, \cdots, a_n$ (for any a_i, we have $0 < a_i \le 1$) and we wish to pack them into as few unit-capacity bins as possible. A unit-capacity bin can accommodate any subset of the a' s whose sum is at most 1. This problem is NP-hard (it is even NP-complete to determine if 2 bins suffice). A simple strategy is to place each a_i in turn into the first bin into which it can fit. Prove that this strategy is a 2-approximation algorithm for bin-packing.

2. (k-Center) The k-Center problem is a classic facility location problem in graph theory and combinatorial optimization. Given a complete undirected graph $G = (V, E)$ where vertices represent points (or cities) and edge weights satisfy the triangle inequality (i.e., the distance metric), the goal is to select k vertices as **centers** such that the maximum distance from any vertex to its nearest center is minimized.

   Formally, let $d(u, v)$ denote the distance (or cost) between vertices u and v. We require $d(u, v) = d(v, u) \ge 0$, $d(u, u) = 0$, and the triangle inequality $d(u, w) \le d(u, v) + d(v, w)$. The problem asks for a set $C \subseteq V$ with $|C| = k$ that minimizes

   $$\max_{v \in V} \min_{c \in C} d(v, c).$$

   The optimal value is called the k-**Center radius** or the **minimum covering radius**.

   Design a $\frac{1}{2}$-approximation algorithm for k-Center problem, and prove the approximation rate of your algorithm.

3. (Weighted Vertex Cover) Consider the **weighted vertex cover problem**: Given an undirected graph $G = (V, E)$ with nonnegative vertex weights $w_v \ge 0$ for each $v \in V$, find a subset $C \subseteq V$ such that every edge $e = (u, v) \in E$ has at least one endpoint in C (i.e., C covers all edges), and the total weight $\sum_{v \in C} w_v$ is minimized.

   (a) Write an **Integer Linear Programming** (ILP) formulation for the problem and its **Linear Programming Relaxation** (LP Relaxation).

   (b) Design a **deterministic rounding approximation algorithm** based on the optimal LP solution. Describe the algorithm steps.

   (c) Prove that your algorithm is a **2-approximation algorithm** (i.e., for every instance, the weight of the solution output is at most twice the optimal value).

   (d) Give a tight example showing that the approximation ratio cannot be better than 2 (i.e., there exists an instance where the algorithm output is exactly twice the optimal solution).